
Uncovering Conceptual Blindspots in Generative Image Models Using Sparse Autoencoders

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Abstract

Generative image models trained on large-scale image-text corpora now synthesize highly realistic images, but they still omit, suppress, or distort concepts that should plausibly be represented in their training data, including human hands and objects appearing in groups of four. Because these failures have largely been documented through anecdotes or narrowly specified probes, the field lacks a systematic account of whether they are isolated prompt-level curiosities or evidence of a broader distributional limitation. We address this gap by formalizing and measuring *conceptual blindspots*: concepts present in natural data that are systematically absent, suppressed, or misrepresented in generated images. The proposed method uses sparse autoencoders (SAEs) to obtain interpretable concept embeddings, thereby enabling quantitative comparison of concept prevalence between real and synthesized images. We train an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts—the largest such SAE to date—and use this representation to analyze conceptual disparities at both distribution and datapoint levels. Applied to four widely used generative models (Stable Diffusion 1.5/2.1, PixArt, and Kandinsky), the framework identifies suppressed blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated blindspots (e.g., wood background texture and palm trees). At the individual datapoint level, the same representation isolates memorization artifacts, including cases in which models reproduce highly specific visual templates seen during training. Together, these results establish a theoretically grounded and empirically scalable framework for evaluating the conceptual fidelity of generative image models relative to an underlying data-generating process.

1 Introduction

Generative image models trained on large-scale datasets now exhibit a level of visual competence that supports applications across computer vision [1-9] and adjacent domains [9-15]. This progress nevertheless coexists with a persistent, empirically documented limitation: qualitative [16-18] and quantitative studies [19-20] show that models can fail on concepts that appear simple from a human perspective, including human hands [21-24], objects appearing in groups of four [25], and negations or object relations [20, 26]. When prompted with such concepts, models often synthesize nearby visual structure rather than the intended concept itself—for example, a hand-like object with six fingers. We refer to these failures as “conceptual blindspots”¹—concepts expected to be represented in the training distribution but absent, suppressed, or distorted in model outputs. The central question is therefore not whether individual prompts can expose surprising errors, but whether those errors instantiate a systematic phenomenon in which generative models under-produce or over-produce broad sets of concepts relative to the natural data distribution.

¹We borrow the term “blindspots” from psychology literature [27], wherein it is used to refer to scenarios where an agent makes biased decisions despite exposure to observations that contradicts their decision’s rationale.



Figure 1: $\delta(k)$ quantifies the relative prevalence of concept c_k in generated images compared with its frequency in natural data. Concepts with $\delta(k) < 0.1$ are classified as **suppressed conceptual blindspots**, whereas concepts with $\delta(k) > 0.9$ are classified as **exaggerated conceptual blindspots**. Images from four popular generative image models illustrate both blindspot regimes and comparatively aligned concepts.

Existing evaluation protocols do not directly answer this concept-level question for generative image models [28, 29]. Widely used metrics such as FID [30] summarize image realism or distributional proximity at a coarse scale, and can therefore obscure failures concentrated in particular concepts. Related measures based on CLIPScore, generation diversity, anomaly detection, or distribution coverage [31–33] provide useful aggregate evidence but do not operate at the granularity of fine-grained visual features or concepts [34, 35]. Conversely, qualitative evaluations such as participant studies [5, 17, 36–38] and open-ended model exploration [39] can reveal concrete failures, but they do not provide a scalable, repeatable, and model-comparable procedure for discovering conceptual blindspots.

This work. We develop an automated, unsupervised methodology for identifying concepts whose prevalence differs between a natural data-generating process and a trained generative image model. The paper contributes a coordinated framework that connects the formal object of study, the technical estimator, and the empirical analysis:

- **Formalizing Conceptual Blindspots in Generative Image Models.** We define conceptual blindspots as concept-level discrepancies between natural and generated images, yielding a quantitative framework for measuring suppression and exaggeration relative to the data-generating process (Section 3). This contribution moves evaluation from anecdotal failures and hand-selected probes to a principled distributional comparison.
- **A Scalable, Unsupervised Approach for Identifying Conceptual Blindspots using Sparse Autoencoders.** We instantiate the formalization with sparse autoencoders (SAEs), which supply a shared, interpretable concept basis for comparing natural and synthesized images (Section 4). We train and open-source an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts, the largest such RA-SAE to date.
- **Exploratory Tool and Analysis.** We provide an exploratory web tool for distribution- and datapoint-level analysis of blindspots across models (Sections 5.1–5.4). Applying the method to Kandinsky, PixArt, and Stable Diffusion (SD) 1.5/2.1 (Section 5), we identify suppressed conceptual blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated conceptual blindspots (e.g., wood background texture and palm trees), as illustrated in Fig. 1.

The model, code, and web tool are available at a software repository.

2 Related Work

2.1 Explainability in Vision

Early explainable-AI work in computer vision emphasized attribution methods that localize influential input regions [40–42]. Although valuable for saliency analysis, these methods provide limited semantic access to learned representations and can produce unreliable explanations [43–45]. Concept-based interpretability [46] addresses this limitation by identifying semantically meaningful directions in neural networks, thereby exposing not only where a model attends but also which concepts and visual structures mediate its behavior [47–49].

Recent analyses show that prominent concept-based methods—ACE [44], CRAFT [48], and Sparse Autoencoders (SAEs) [50, 51]—can be understood as variants of a dictionary-learning problem with

different constraints [52]. Within this family, SAEs provide a scalable mechanism for extracting sparse, interpretable concept activations. Although standard SAEs have known limitations, including overly specific features [53], compositionality challenges [54], and limited intervention effects [55], improved variants such as archetypal SAE (RA-SAE) [56] are designed to yield more stable and data-faithful dictionaries.

Other interpretability tools, including prompt-based probing [57], attention or activation visualizations [58, 59], and dataset-level statistics [32, 33], provide complementary but partial evidence. They generally operate on full images, prompts, or aggregate coverage measures rather than on fine-grained features and concepts [34, 35]. They also often require subjective interpretation and do not separate distinct failure modes [60]. These limitations motivate an explicitly concept-level diagnostic for generative image models [28], one that can preserve semantic granularity while remaining scalable across many concepts and model families.

2.2 Generative Image Models

Diffusion-based methods now dominate generative vision modeling across images [4, 61–64], video [65–72], and 3D generation [7, 73–75]. In image generation, this trajectory begins with denoising diffusion probabilistic models (DDPMs) [76] and extends through non-Markovian denoising diffusion implicit models (DDIMs) [77].

The Stable Diffusion (SD) [9] family made diffusion models broadly accessible to research and open-source communities. The original SD model was followed by SD 2 [78], SD 3 [79], and SD XL [80], together with extensions that condition generation on style [81, 82], pose [83], identity [84, 85], and alternative input modalities such as image-to-image generation. Additional latent diffusion systems, including Kandinsky [86], PixArt [87], DeepFloyd [4, 88], and FLUX [89], expand the architectural landscape for which concept-level comparisons are scientifically relevant.

2.3 Datasets for Generative Image Models

The performance of contemporary generative vision models depends substantially on computational scale and large internet-derived datasets [90–92]. LAION-5B [93], collected from Common Crawl [94], has been especially influential for open-source text-to-image systems, including SD and its derivatives; it contains more than 5B image-caption pairs, of which 2.3B are in English. Other widely used resources include COYO-700M [95] and Conceptual Captions [96], with 700M and 3M image-caption pairs, respectively.

The adoption of LAION-5B also intensified concerns about bias and unsafe content [97–101]. Despite filtering, harmful material remained [97–99], including NSFW images [100] and hundreds of CSAM instances [101], prompting temporary removal from official channels. LAION-5B further contains low-quality images [102] and internet-style captions, such as product descriptions, that can differ from user prompts supplied to trained models [103]. These properties make concept-level comparison between natural and generated images especially important: aggregate quality metrics may miss systematic distortions inherited from, or amplified relative to, the training distribution.

2.4 Concept Discovery and Sparse Coding in Generative Image Models

Dictionary learning constructs sparse representations in which each sample is reconstructed from a small number of dictionary atoms [104, 106]. Building on compressed sensing theory [107, 108], the area has progressed from vector quantization [109] to methods such as Non-negative Matrix Factorization [110, 111], Sparse PCA [112], and K-SVD [113]. More recent developments include online learning [114], structured sparsity [115], theoretical recovery guarantees [116, 117], and increasingly direct connections to deep representation learning [118, 119].

Sparse coding has also been used to study the emergence of high-level concepts inside diffusion models [120]. Before diffusion models, concept-grounded interpretability for generative systems relied on concept-bottleneck models, which require human intervention during training [121], or post-hoc detectors that retrofit concept supervision [122]. Because both strategies depend on human-defined concepts, they can miss systematic concept-level trends that are not specified in advance.

Aggregate metrics, including precision, recall, density, and coverage [35, 123], as well as latent density scores that estimate sample quality from the model’s latent space [124], are useful for

measuring global generative behavior. They are less suited to identifying which specific concepts are under- or over-represented. This gap motivates a scalable, unsupervised framework that quantifies concept-level failure modes in generative image models [125].

3 Formalizing Conceptual Blindspots in Generative Models

We formalize *conceptual blindspots* as systematic discrepancies between the concepts expressed in natural images and those expressed in model-generated images. The formalization produces quantitative measures of concepts that are suppressed or exaggerated by a generative model relative to the data distribution. The construction is summarized in Fig. 2. Although the definitions rely on standard assumptions from representation learning and identifiability [126–130], our empirical analysis shows that the resulting measurements expose meaningful model behavior even when these assumptions hold only approximately.

Definition 1 (Data-Generating Process). Let $\mathcal{C} \subset \mathbb{R}^K$ denote a latent space with a Boltzmann prior $p(\mathbf{c}) = \exp(-E(\mathbf{c}))Z^{-1}$, where $E(\cdot)$ denotes an energy function that linearly decomposes over individual latents and Z is the corresponding partition function, i.e., $E(\mathbf{c}) = \sum_k E(c_k)$ and hence $p(\mathbf{c}) = \prod_k p_k(c_k)$. A data-generating process (DGP) is an invertible function $G : \mathcal{C} \rightarrow \mathcal{X}$ that maps the latents $\mathbf{c} \in \mathcal{C}$ to observations $\mathbf{x} \in \mathcal{X}$, i.e., $\mathbf{x} = G(\mathbf{c})$.

For notational simplicity, we use $p(\cdot)$ to denote both the latent density $p(\mathbf{c})$ and its pushforward to image space $p(\mathbf{x})$, where $\mathbf{x} = G(\mathbf{c})$. This convention is justified by the invertibility of G , which induces a valid distribution over $\mathcal{X}$ via the change-of-variables formula. Accordingly, the coordinates of the latent space define the **Concepts** that organize the image distribution $P_{\mathcal{X}}$ over the observation space $\mathcal{X}$. Such latents may encode attributes including color, shape, size, and location [131][132]. We further allow the data-generating process to associate each image with a text description $\mathbf{t} \in \mathcal{T}$, while leaving the captioning mechanism implicit. A text-conditioned **Generative image model** g_{θ} with parameters θ is then trained on image–text pairs to map a noise signal $\eta \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and a scene description $\mathbf{t}$ to an image $\mathbf{x}$ depicting that description.

To identify blindspots in g_{θ} , we compare the probability mass assigned to each concept by the data-generating process with the corresponding probability mass induced by the model. Starting from an evaluation set of natural image–text pairs $(\mathcal{D}_{\mathcal{X}}, \mathcal{D}_{\mathcal{T}})$, we construct a synthetic set $\mathcal{D}'_{\mathcal{X}}$ by generating images from the same text descriptions. Concept occurrence is estimated with an **Energy model** $\xi : \mathcal{X} \rightarrow \mathbb{R}^d$ that maps each image to a d -dimensional vector whose k^{th} coordinate represents the energy of concept k , i.e., $\xi_k(\mathbf{x}) = E(c_k)$. The invertibility assumption on the data-generating process makes these estimates well defined. The probability associated with an image $\mathbf{x}$ can then be expressed as $p_k(\mathbf{x}) \propto \exp(-\sum_k \xi_k(\mathbf{x}))$ (where Z_{ξ} is the partition function), yielding the population-level estimate $p_k(\mathcal{D}_{\mathcal{X}}) = \prod_{\mathbf{x} \in \mathcal{D}_{\mathcal{X}}} p_k(\mathbf{x})$. Using this construction and the sigmoid map $\sigma(\cdot)$, we define the average energy difference for concept k between $\mathcal{D}_{\mathcal{X}}$ and $\mathcal{D}'_{\mathcal{X}}$.

Definition 2 (Energy Difference). Let $\mathbf{x} \in \mathcal{D}_{\mathcal{X}}$ denote a real image sampled from the data-generating process G , and let $\mathbf{x}' \in \mathcal{D}'_{\mathcal{X}}$ be a synthetic image generated by the model g_{θ} . Let $\xi_k : \mathcal{X} \rightarrow \mathbb{R}$ denote the energy assigned to the k^{th} concept by the energy model ξ . We define the energy difference for concept k as:

$$\begin{aligned} \delta_{g_{\theta} \leftrightarrow G}(k) &= \sigma(\mathbb{E}_{\mathbf{x}'}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x}}[\xi_k(\mathbf{x})]) \\ &= \frac{p_k(\mathcal{D}'_{\mathcal{X}})}{p_k(\mathcal{D}_{\mathcal{X}}) + p_k(\mathcal{D}'_{\mathcal{X}})}, \end{aligned} \quad (1)$$

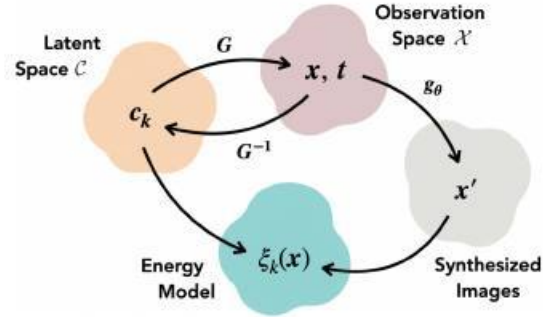


Figure 2: **Intuitive mapping of our framework.** Latent concepts $c_k \in \mathcal{C}$ are mapped to observations $(\mathbf{x}, \mathbf{t})$ through the (invertible) DGP. g_{θ} generates images $\mathbf{x}'$ from $\mathbf{t}$. The energy model $\xi_k(\mathbf{x})$ extracts concept representations from both $\mathbf{x}$ and $\mathbf{x}'$, enabling concept-distribution comparisons that identify blindspots.

where the expectations are taken over $\mathcal{D}'_{\mathcal{X}}$ and $\mathcal{D}_{\mathcal{X}}$, respectively, and $p_k(\mathcal{D}) \propto \exp(-\sum_{\mathbf{x} \in \mathcal{D}} \xi_k(\mathbf{x}))$ denotes the unnormalized conceptual probability mass of dataset $\mathcal{D}$ under concept k .

Thus, the energy difference for concept k expresses the relative probability of observing that concept in a target set of observations (here, $\mathcal{D}'_{\mathcal{X}}$) compared with a baseline set (here, $\mathcal{D}_{\mathcal{X}}$). This statistic directly induces the following definition of conceptual blindspots.

Definition 3 (Suppressed / Exaggerated Conceptual Blindspots). Given a generative image model g_{θ} , we say, compared to the data-generating process G , c_k is a suppressed conceptual blindspot in the model if $\delta_{g_{\theta} \leftrightarrow G}(k) < \lambda_{\min}$ and exaggerated if $\delta_{g_{\theta} \leftrightarrow G}(k) > \lambda_{\max}$.

A conceptual blindspot is therefore a concept whose likelihood in generated images deviates markedly from its prevalence under the data-generating process, either through suppression or exaggeration. Suppressed concepts have disproportionately low activation (e.g., $\delta(k) < \lambda_{\min}$), whereas exaggerated concepts are overrepresented (e.g., $\delta(k) > \lambda_{\max}$). Throughout the empirical analysis, we use $\lambda_{\min} = 0.1$ and $\lambda_{\max} = 0.9$ to isolate the two regimes.

This definition is related to “mode collapse” in prior work (e.g., see Bau et al. [39]), but it shifts the unit of analysis. Mode collapse concerns the exaggerated or suppressed probability of generating *entire images*; conceptual blindspots instead concern altered probabilities of specific concepts within images. For instance, if a model fails to generate objects with a `white background`, the missing background attribute is a suppressed conceptual blindspot.

4 Method: Operationalizing the Definition of Conceptual Blindspots

We now instantiate the formal definition as a concrete pipeline for detecting conceptual blindspots in a generative model g_{θ} . Sec. 3 requires two empirical objects: (i) generated images sampled from g_{θ} under captions that permit comparison with the natural data-generating process, and (ii) an energy model that maps both natural and generated images into a common concept basis. We use $\|\cdot\|_F$ for the Frobenius norm and $\|\cdot\|_0$ for the number of nonzero entries (the ℓ_0 pseudo-norm). For a vector or matrix $\mathbf{X}$, $\mathbf{X} \geq 0$ denotes element-wise non-negativity. For $n > 0$, $[n] := \{1, \dots, n\}$, and the i -th row of a matrix $\mathbf{A}$ is denoted by $\mathbf{A}_i$.

From Prompts to Latent Representations. To identify conceptual blindspots in g_{θ} , we compare natural image–caption pairs $(\mathbf{x}, t)$ sampled from the data-generating process G with synthetic counterparts generated by g_{θ} from the same text descriptions.

Given a caption t , we synthesize the paired image $\mathbf{x}'$ with a pretrained text-to-image generator $g_{\theta} : \mathcal{T} \rightarrow \mathcal{X}$ implemented as a denoising diffusion probabilistic model (DDPM) [78, 86, 87]. Sampling proceeds in latent space through the reverse trajectory $(\gamma_t)_{t=0}^T$:

$$\gamma_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \gamma_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\gamma_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \epsilon_{\theta}(\gamma_t, t, c) \right) + \sigma_t \eta_t, \quad \text{and} \quad \eta_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

where $\alpha_t \in (0, 1)$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ follow the standard cosine noise schedule. The final latent γ_0 is decoded by a pretrained VAE to obtain $\mathbf{x}' = \text{VAE}(\gamma_0)$. In all subsequent analyses, g_{θ} is treated as a black-box prompt-to-image map $t \mapsto \mathbf{x}'$.

Defining the Energy Model. Prior work shows that self-supervised learning can recover structure related to the data-generating process and identify the underlying energy function up to linear transformations [126, 128, 133, 134]. We therefore use DINOv2 [135], a state-of-the-art self-supervised vision model, as the feature extractor for our energy model. Because the number of concepts underlying the DGP may exceed the dimensionality of the model feature space [51, 136], we train sparse autoencoders (SAEs) on DINOv2 features to recover subspaces associated with these concepts [50, 56, 137–139]. The operational assumption is that if DINOv2 represents concepts through approximately orthogonal directions, as reflected by the independence constraint in Def. 1, then the SAE can separate those concepts into individual latent dimensions [136]. The activation assigned by the SAE to a dimension then serves as our empirical approximation of the ground-truth energy function for the concept modeled by that dimension.

Formally, let $\mathbf{f} : \mathcal{X} \rightarrow \mathbb{R}^d$ denote the DINOv2 feature extractor. For each natural or synthetic image $\mathbf{x}$ in $\mathcal{D}_{\mathcal{X}}$ or $\mathcal{D}'_{\mathcal{X}}$, we compute $\mathbf{a} = \mathbf{f}(\mathbf{x}) \in \mathbb{R}^d$. For n image–text pairs, the natural and generated

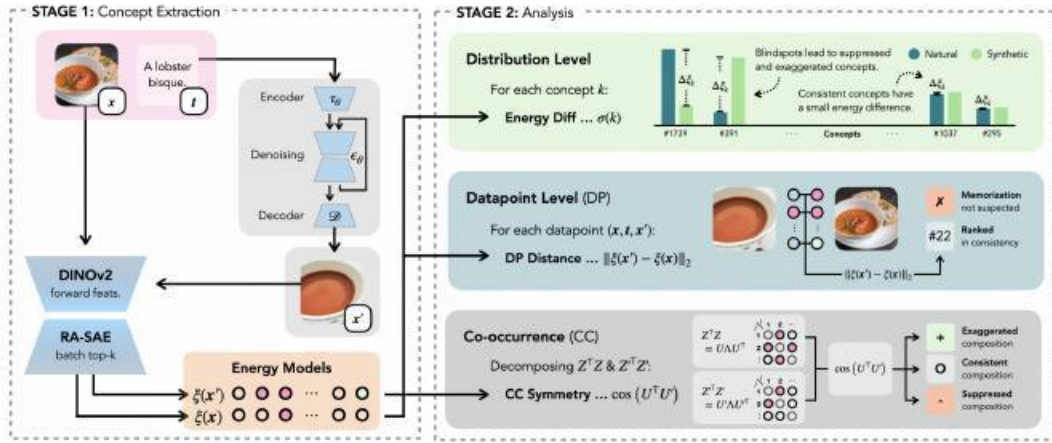


Figure 3: **Concept Extraction Pipeline.** For a triplet (x, t, x') , each image is embedded with DINOv2 and processed by a RA-SAE into sparse concept embeddings, yielding energy models $\xi(x)$ and $\xi(x')$, respectively. In particular, $\xi_k(x) = \xi_k(x')$ (Each datapoint vector is decomposed by an SAE into a sparse combination of learned concept atoms. Let $D \in \mathbb{R}^{d \times K}$ be a dictionary of K' concept vectors, and let $\Psi : \mathbb{R}^d \rightarrow \mathbb{R}^{K'}$ be the encoder that maps features to sparse concept codes. Applying Ψ row-wise produces the activation matrix $Z = \Psi(A) \in \mathbb{R}^{n \times K'}$, with each row $z_i = \Psi(a_i)$ representing the concept decomposition of one image. The decoder reconstructs features as ZD^T , and the SAE objective minimizes reconstruction error subject to sparsity and non-negativity:

$$\min_{\Psi, D} \|A - \Psi(A)D^T\|_F^2 \quad \text{s.t.} \quad \Psi(A) \geq 0, \|\Psi(A)_i\|_0 \ll K' \quad \forall i \in [n]. \quad (2)$$

Standard SAEs can converge to arbitrarily oriented dictionaries, making downstream concept analyses sensitive to random initialization. To obtain a reproducible and data-grounded concept basis, we use the *Archetypal* SAE (RA-SAE) [56] with a TOP-K sparsity constraint [137]. RA-SAE constrains the dictionary D to be a convex combination of training examples. Specifically, $D = WA$ with $W \in \Omega_{K', n}$, where $\Omega_{K', n}$ is the set of row-stochastic matrices in $\mathbb{R}^{K' \times n}$:

$$\Omega_{K', n} := \{W \in \mathbb{R}^{K' \times n} \mid W \geq 0, W\mathbf{1} = \mathbf{1}\}. \quad (3)$$

Consequently, every atom D_i lies in the convex hull of the data $\text{conv}(A)$, and every reconstruction ZD^T lies in the conic hull of the data $\text{cone}(A)$. This constraint anchors learned concepts to the support of the data distribution [56]. After training, the SAE yields aligned sparse codes Z for natural images and Z' for their generated counterparts. *These codes capture the same prompt-conditioned visual semantics in terms of shared, interpretable concepts, with the activation value of the concept serving as energy values for our analysis of conceptual blindspots.* Thus, for each prompt and associated natural image (t, x) , the pipeline synthesizes x' and maps both images into a shared sparse concept space, producing (z, z') for direct comparison of natural and generated visual content.

This pipeline operationalizes conceptual fidelity: a captioned image (t, x) and its generated counterpart x' are encoded by DINOv2 and RA-SAE into interpretable concept representations (z, z') , which provide the empirical basis for measuring concept shifts induced by the generative process.

5 Results

We evaluate four generative image models trained on LAION-5B or related derivatives—SD 1.5, SD 2.1, PixArt, and Kandinsky—using $|\mathcal{X}| = 10,000$ image-text pairs and their corresponding generations (Fig. 4). The analysis is organized at three complementary levels (Fig. 3): a ● **distribution-level** evaluation identifies suppressed and exaggerated concepts; a ● **datapoint-level** analysis localizes failures associated with ambiguity, omission, and memorization; and a ● **compositional** analysis characterizes distortions in concept co-occurrence geometry.



Figure 4: **Representative image-caption pairs** from LAION-5B and prompt-matched generations from SD 1.5, SD 2.1, PixArt, and Kandinsky.

The empirical study is supported by an interactive exploratory tool, shown in Fig. 5. The web interface centers on UMAP projections of concept representations, enabling direct visualization and comparison of concept-level energy differences. It is publicly available at a software repository, together with pre-computed energy-difference data for the four evaluated models (SD 1.5, SD 2.1, PixArt, and Kandinsky). The analyses below use this interface to move systematically between global distributional patterns and individual examples. Its primary functionalities are:

- **Contrast different models and architectures.** For each evaluated model, the tool provides a UMAP visualization over all 32,000 RA-SAE concepts. Each point denotes one concept and is color-coded by its energy difference.
- **Inspect concepts.** Each concept card reports key statistics, representative real and generated images (x , x'), and visualized co-occurrence patterns.
- **Explore blindspots.** Beyond UMAP and per-concept views, the tool provides global rankings of suppressed and exaggerated blindspots, making the most pronounced concept-level deviations directly inspectable.

Rather than exhaustively analyzing a single failure mode, we use the tool to organize high-level findings that demonstrate how the framework supports customizable concept-level investigation.

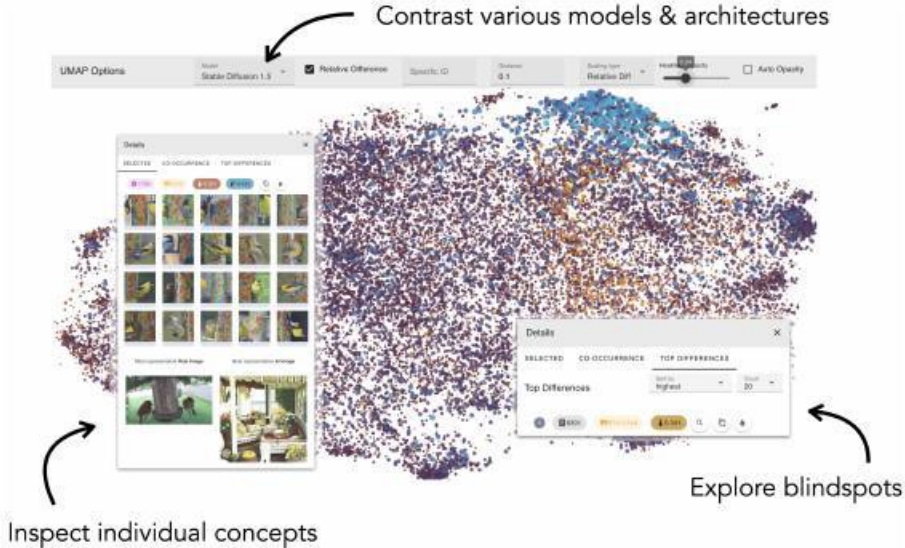


Figure 5: **Overview of the Exploratory Tool.** The web interface presents a UMAP projection for each evaluated model, with each dot corresponding to one concept and color indicating its energy difference. Selecting a concept opens a detail panel containing illustrative images, statistics, representative natural and generated images x and x' , an ordered list of co-occurrences, and global rankings of blindspots.

5.1 Generative Image Models Cannot Produce Unseen Concepts

To measure concept-level disparities between generated and natural images, we first analyze the marginal energy difference $\delta(k)$ for the 32,000 RA-SAE concepts. As defined in Sec. 4, $\delta(k)$ measures the relative prevalence of concept k in synthesized images compared with natural images. Values $\delta(k) < 0.1$ indicate under-representation, or suppression, whereas values $\delta(k) > 0.9$ indicate over-representation, or exaggeration. Fig. 6 reports the resulting distributions for the four evaluated models. All four models exhibit heavy-tailed histograms with substantial mass near both extremes, demonstrating that concept coverage errors are not confined to isolated cases. The left tail, corresponding to suppressed concepts, is consistently denser and longer than the right tail, indicating a stronger tendency toward omission than exaggeration. This asymmetry is quantified by negative

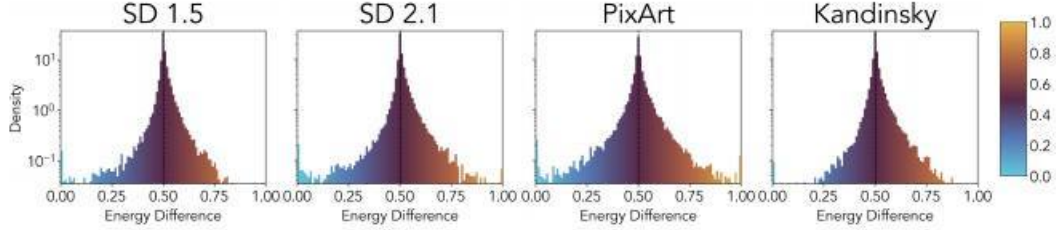


Figure 6: **Concept Energy Distribution.** Log-scale histograms of energy differences $\delta(k)$ over 32,000 concepts for each evaluated model, comparing concept prevalence in natural and synthesized images. Values left of zero indicate suppressed concepts; values right of zero indicate exaggerated concepts.

skewness in every model: Skewness = -0.54 for SD 2.1, -0.40 for both SD 1.5 and PixArt, and -0.23 for Kandinsky.

Although all four models contain both suppressed and exaggerated concepts, their profiles are not identical. PixArt, for example, exhibits a wider spread, consistent with a more dispersed suppressed-concept distribution. The shared left-skew establishes concept omission as a recurring property across the evaluated models, while the model-specific distributional shapes motivate the structural analysis in the following sections.

5.2 Structure and Specificity of Conceptual Blindspots

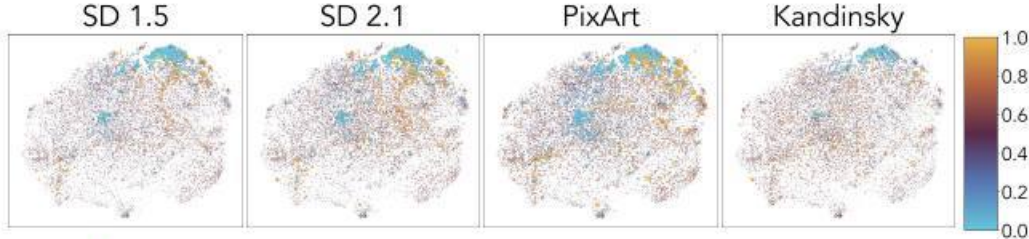


Figure 7: **Structure of Concept Energy Differences.** UMAP visualizations of 32,000 concepts, colored by energy difference $\delta(k)$ between natural and synthesized images. Clustered regions reveal structured conceptual blindspots, with suppressed concepts shown in blue and exaggerated concepts in yellow.

Having established marginal concept-frequency discrepancies, we next examine whether blindspots possess coherent organization in concept space. We embed all 32,000 concepts into two dimensions with UMAP applied to the sparse codes and color each concept by $\delta(\cdot)$. Fig. 7 shows that blindspots form distinct clusters across all models. These clusters often correspond to contiguous regions of suppressed concepts, indicating that blindspots are structured rather than randomly distributed and may reflect shared properties of the training data or architectural priors. We then quantify the cross-model consistency of these structures by comparing concept-level $\delta(\cdot)$ vectors. Fig. 8 reports scatter plots and pairwise Pearson correlations between SD 1.5 and the other models. SD 1.5 and SD 2.1 are strongly correlated ($r = 0.82$), consistent with their related architectures and training pipelines. Correlations with PixArt and Kandinsky are lower— $r = 0.41$ and $r = 0.46$, respectively—showing that different model families concentrate their blindspots in different regions of the concept space.

Together, these results establish that conceptual blindspots contain both shared and model-specific components. Shared blindspots plausibly reflect properties of the data distribution, whereas model-specific blindspots point to differences in architecture, training dynamics, or capacity; the qualitative analysis below therefore examines both universal and unique cases.

5.3 Qualitative Examples of Blindspots

We next visualize specific suppressed and exaggerated blindspots to clarify the kinds of concepts captured by each regime. Specifically, Fig. 9a shows a conceptual blindspot suppressed by all evaluated models: each model fails to reproduce the concept `solid white` on documents. Despite

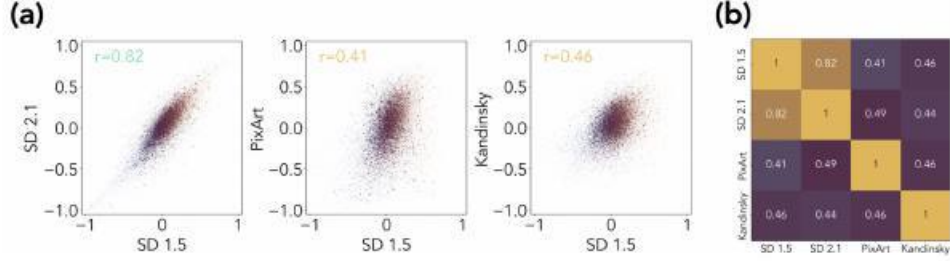


Figure 8: **Cross-Model Concept Energy Correlation.** Pairwise scatter plots of $\delta(k)$ across all four evaluated models, with Pearson correlation coefficients reported in the upper left. Strong alignment between SD 1.5 and SD 2.1 contrasts with weaker correlations involving other architectures, indicating model-specific blindspot profiles. **Correlation Matrix of Conceptual Blindspots.** Heatmap of pairwise Pearson correlation coefficients for δ between all models, quantifying how much conceptual blindspot structure is shared across models.

captions that explicitly reference this concept, none of the generated images reflects the intended visual semantics, suggesting that this region of the concept space is systematically under-sampled across models. Fig. 9b, by contrast, highlights a model-specific blindspot: the concept pan is accurately captured by three models but conspicuously absent from generations produced by Kandinsky. This qualitative contrast reinforces the finding from Sec. 5.2 that cross-model agreement is high for some blindspots but limited for others.

Conversely, Fig. 10 presents an exaggerated blindspot in which the concept shadow under animal is over-emphasized in generated images. Although shadows are visually plausible in these scenes, their consistent and pronounced rendering across models, relative to the more nuanced and variable occurrences in natural images, indicates an overactive prior. Despite attempts to identify concepts uniquely exaggerated by a specific model, we did not find clear examples, suggesting that the observed exaggerations are approximately universal within the evaluated set.

Overall, these examples show how conceptual blindspots manifest in generated outputs and demonstrate that the energy-based diagnostic surfaces both shared and model-specific failure modes. The consistent failure to reproduce clear or solid background elements, such as whitespaces on documents, is especially informative because it suggests that some failure patterns may arise from architectural constraints or training-data biases that transcend individual model idiosyncrasies.

While these aggregate-level analyses are informative, they mo-

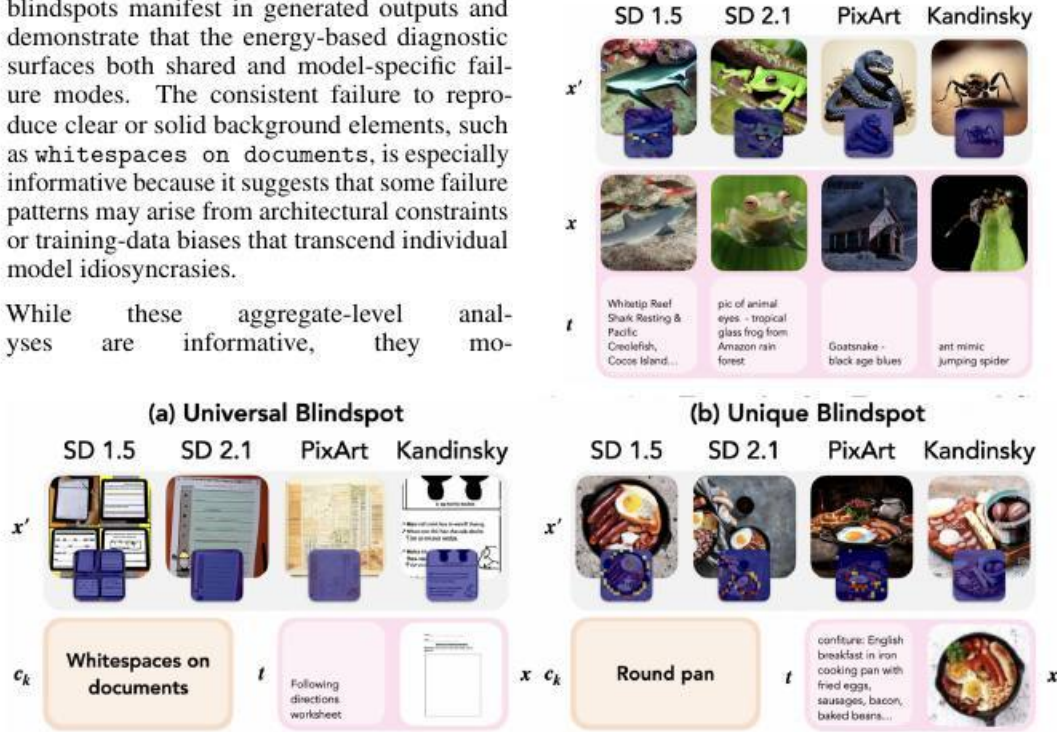


Figure 9: **Examples of Suppressed Conceptual Blindspots.** The natural images x , representative c_k and t , shown alongside four synthesized images x' , generated using S_θ . The universal blindspot is present in all evaluated models; the unique blindspot is only present in Kandinsky.

340 tivate a finer-grained ques-
 341 tion: do blindspots appear
 342 only as population-level dis-
 343 crepancies, or do they also
 344 manifest at individual dat-
 345 apoints? This perspective
 346 probes blindspot mechanisms
 347 more directly by exposing
 348 cases of prompt misinterpreta-
 349 tion, latent memorization, or
 350 both.
 351

352 5.4 Investigating Datapoint-level Energy Difference: From Incongruent to Memorized Images

353 To move beyond population-level statistics, we examine individual natural vs. generated image pairs
 354 for which the $\delta(\cdot)$ values averaged across all concepts exhibit the largest and smallest differences.
 355 This analysis makes model successes and fail-
 356 ures interpretable at the example level; the latter
 357 often arise from prompt ambiguity or memo-
 358 rization artifacts. For example, Fig. 11a shows
 359 instances with near-zero difference in average
 360 $\delta(\cdot)$ values. In these cases, the generated im-
 361 ages are conceptually nearly indistinguishable
 362 from the originals. Qualitative inspection in-
 363 dicates that this agreement does not necessar-
 364 ily reflect faithful prompt semantics; instead,
 365 it can arise from replication of memorized tem-
 366 plates, including repetitive visual structures such
 367 as clothing outlines or object arrangements that
 368 may correspond to frequent training patterns.
 369 By contrast, Fig. 11b illustrates samples that
 370 are among the largest $\delta(\cdot)$ values. These sam-
 371 ples indicate substantial conceptual divergence
 372 between the synthesized and natural image. Al-
 373 though some discrepancies can be attributed to
 374 underspecified or noisy captions, others reveal
 375 genuine blindspots: the prompt describes a clear
 376 concept faithfully present in $\mathcal{D}_X$, yet the model fails to realize it in $\mathcal{D}'_X$. This failure suggests that,
 377 even when language grounding is adequate, certain concepts remain outside the model’s effective
 378 generative repertoire.



Figure 11: **Datapoint-level Conceptual Alignment.** (a) Examples with minimal energy differences where models appear to memorize training patterns. (b) Examples with large differences where significant concept divergences due to prompt ambiguity or model limitations occur.

379 5.5 Analyzing Post-Training Effects

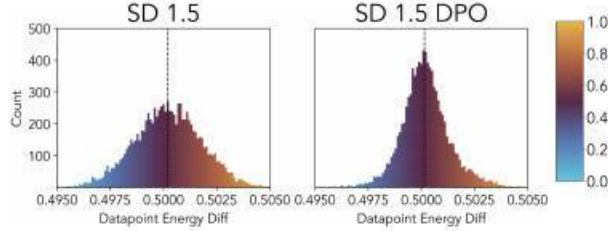


Figure 12: **Effect of DPO on Concept Fidelity.** Histogram of datapoint-wise energy differences between the synthesized and natural distribution of SD 1.5 models with and without DPO.

380 Post-training protocols, including safety fine-tuning, have been argued to reduce the diversity of
 381 model generations [140]. Given the pipeline’s ability to isolate concept-level differences between
 382 model generations and the ground-truth DGP, we next use it to characterize the effects of DPO—a
 383 popular safety fine-tuning protocol [141]. Specifically, we compare two variants of SD 1.5: one
 384 trained with DPO and one without DPO. For each image pair $(\mathcal{D}_X, \mathcal{D}'_X)$, we compute the ℓ_2 norm of
 385 the difference between their internal concept energy vectors, $\|\xi(\mathcal{D}'_X) - \xi(\mathcal{D}_X)\|_2$. Fig. 12 presents
 386 a histogram of these datapoint-wise energy differences. The DPO-enhanced model exhibits both a
 387 lower median and a narrower spread, indicating a more consistent distribution of generated concepts
 388 relative to the ground-truth DGP. This pattern suggests that DPO may regularize the model’s concept
 389 distribution, encouraging outputs that more closely reflect the semantic content of the observed inputs.
 390 While our analysis does not disentangle the specific inductive biases introduced by DPO, these results
 391 provide empirical evidence that its optimization objective, which favors human-preferred generations,
 392 indirectly promotes closer alignment with the training distribution. In particular, it reduces *both over*
 393 *and under* activation of individual concepts relative to the natural baseline. These findings highlight
 394 the utility of the pipeline for characterizing downstream post-training effects not only in terms of
 395 output quality, but also in terms of how they reshape the conceptual geometry of the model’s output
 396 space.

397 5.6 Conceptual Misalignment as a Function of Empirical Frequency

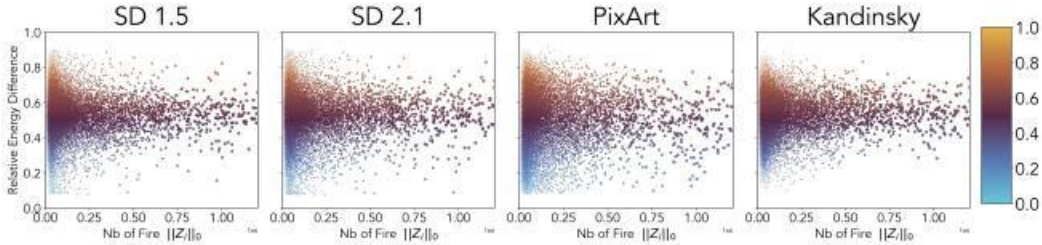


Figure 13: **Concept Fidelity Across Frequency Spectrum.** Scatter plots show the relationship between concept frequency (x-axis) and energy difference (y-axis) across the four evaluated models. Each point represents one concept, with marker size proportional to its activation frequency.

398 We next test whether conceptual blindspots are not merely architectural artifacts, but also emerge
 399 from distributional peculiarities of particular features. We examine whether concepts that are rarely
 400 activated in natural images—the long tail of the data distribution—are more likely to become
 401 blindspots in generative models. Specifically, we process the natural dataset $\mathcal{D}_X$ through the trained
 402 SAE and compute, for each concept k , its empirical frequency $\|\mathbf{Z}_{:,k}\|_0$, where $\mathbf{Z}_{:,k}$ denotes the
 403 activations of concept k across all images. We then correlate this frequency with the absolute energy
 404 difference observed across generated outputs. Fig. 13 visualizes this relationship for all evaluated
 405 models. Concepts with higher frequency in natural data tend to show lower energy discrepancies,
 406 whereas rare concepts—especially suppressed ones ($\delta(k) < 0.5$)—exhibit substantial alignment
 407 errors. This pattern indicates that many blindspots arise not from randomness or isolated model
 408 quirks, but from systematic effects tied to long-tail concept distributions. Addressing these issues may
 409 therefore require not only architectural changes, but also data-centric strategies such as reweighting
 410 or augmentation.

6 Discussion

Our analysis reveals multiple conceptual blindspots in four popular generative image models. The results presented here, however, only scratch the surface: each individual finding could warrant its own dedicated investigation. Rather than delving deeply into any one of these questions, we instead showcase the versatility of our method and exploratory tool. Out of the box, they allow for a systematic identification of concepts that models struggle to generate, detection of memorization artifacts, discovery of datapoints with insufficient captions, quantification of post-training effects, and characterization of conceptual shifts across model architectures. We thus open space for follow-up work to extend the depth of analysis, scope of evaluated architectures, and inquiry into root causes of conceptual blindspots. Future work could also explore hierarchical representations of concepts in RA-SAE to allow for a more nuanced analysis.

Limitations. We wish to highlight several limitations of our work. By relying on DINOv2 and RA-SAE for concept extraction and representation, our approach is inherently constrained to the kinds of concepts these models capture; concepts poorly represented by them will escape our analysis. Additionally, while our sample size of 10,000 images is substantial, it may not fully capture the long tail of rare concepts, concept co-occurrence, or other compositional statistics.

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Appendix

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A Experimental Setup

This section details the experimental setup for our analysis of four popular generative image models: SD 1.5/2.1, Kandinsky, and PixArt, all trained on LAION-5B or its subsets/derivatives. The code is available at a software repository.

A.1 Observation Space

The observation space is constructed by sampling 10,000 image-text pairs from the LAION-5B dataset [93], which serves as our domain of natural images. Due to concerns with CSAM and other unsafe content in the dataset, the original data release is no longer available. A substitute release of a subset of this dataset with additional filtering of the unsafe content, available at <https://huggingface.co/datasets/laion/relaion2B-en-research-safe> is used.

The sampling procedure consists of: (1) loading the full LAION dataset using the Hugging Face `datasets` library, (2) performing validation to ensure proper URL structure and resource availability via HTTP HEAD requests, and (3) employing random sampling with replacement until reaching the target count of 10,000 valid samples. This approach, yielding D_G with (x, t) tuples, ensures our observation space contains accessible image-text pairs for comparative analysis of a dataset of image URLs whose large portion has been made unavailable since original release. Additional examples of synthesized images are shown in Appendix H.

A.2 Synthesized Images

For each of the four evaluated models, we generate a synthetic dataset D_{g_θ} to have a one-to-one correspondence with D_G , yielding triplets (x, x', t) . Specifically, given the 10,000 image-text pairs (x, t) from D_G , we use t to synthesize counterpart images x' using each generative model g_θ .

The synthesis process follows the standard text-to-image generation pipeline for each model architecture, implemented using the Hugging Face `diffusers` library, where the models are loaded at mixed precision (fp16). All synthetic images are generated at 512×512 pixel resolution with default parameters.

Stable Diffusion 1.5. The checkpoint from <https://huggingface.co/benjamin-paine/stable-diffusion-v1-5> (which is a mirror of the deprecated <https://huggingface.co/runwayml/stable-diffusion-v1-5>) is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Stable Diffusion 1.5 + DPO. The DPO variant of SD 1.5 (used in the analysis in Sec. 5.5) follows the baseline SD 1.5 implementation, but replaces the UNet component with a DPO-trained version from <https://huggingface.co/mhdang/dpo-sd1.5-text2image-v1>.

Stable Diffusion 2.1. The checkpoint from <https://huggingface.co/stabilityai/stable-diffusion-2-1> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Kandinsky. The checkpoint from <https://huggingface.co/kandinsky-community/kandinsky-2-1> is used. Inference is performed using 100 inference steps, with the guidance scale fixed at 4.0.

PixArt. The checkpoint from <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

A.3 Distribution Level Analysis

Section 5.1. We compute energy differences $\delta(\cdot)$ across all 32,000 concepts for each evaluated model. The sigmoid transformation with temperature $T = 0.8$ is applied during normalization. The resulting values are visualized as log-scale density histograms with 100 bins spanning $[0, 1]$.

Section 5.2 We embed the complete set of 32,000 concepts into two-dimensional space using UMAP applied to the sparse concept codes. Each point in this UMAP represents an individual concept, colored according to its energy difference $\delta(\cdot)$, emphasizing both suppressed and exaggerated blindspots. To quantify cross-model consistency, we compute pairwise Pearson correlation coefficients between $\delta(\cdot)$ vectors of all model pairs, producing both scatter plots and correlation matrices. This analysis reveals whether blindspots cluster in conceptual space and identifies model-specific versus universal patterns of conceptual blindspots.

Section 5.3 We rank all 32,000 concepts by their energy difference $\delta(\cdot)$, and manually examine the extrema (both suppressed and exaggerated blindspots). For suppressed blindspots, we select concepts with $\delta(\cdot) < 0.1$; for exaggerated blindspots, we choose those with $\delta(\cdot) < 0.9$. Presented examples are manually annotated with textual descriptions of the respective concepts through inspection of their most activating images and spatial attention patterns. We outline ongoing efforts to automate this concept interpretation in Appendix D

Section 5.5 We compare 1.5 with and without DPO in the following fashion: for each image pair (x, x') , we compute the L2 norm of the difference between their concept energy vectors $\|\xi(x') - \xi(x)\|_2$. We apply a sigmoid transformation with temperature $T = 0.8$ to the element-wise differences before taking their mean. This yields datapoint-wise energy differences that quantify how much each generated image deviates from its natural counterpart in concept space. Finally, these differences are visualized as overlapping histograms, contrasting both model variants.

Section 5.6 For each concept c_k , its empirical frequency $\|\mathbf{Z}_{:,i}\|_0$ (the count of non-zero activations across the natural dataset) is counted. A sigmoid normalization with temperature $T = 0.4$ is then applied to the energy differences $\delta(\cdot)$. The analysis is visualized using scatter plots where the x-axis is the empirical concept frequency and the y-axis is the sigmoid-transformed energy difference. The point sizes are proportional to activation frequency and point colors are proportional to the magnitude of energy differences.

A.4 • Datapoint Level Analysis

Section 5.4 For each image pair (x, x') , we compute the L2 norm of the difference between their concept energy vectors $\|\xi(x') - \xi(x)\|_2$. This yields a scalar measure of conceptual divergence for each image pair. The samples are ranked by their energy differences. Minimal divergence indicate potential memorization artifacts and maximal divergence point to significant conceptual failures. This analysis enables qualitative inspection of specific failure modes.

A.5 • Co-occurrence Analysis

Appendix F For both the natural and synthesized data D_G and $D_{g\theta}$, concept co-occurrence patterns are analyzed through the co-activation matrix $Z^T Z$, which holds pair-wise correlations in concept usage. Spectral analysis is performed to examine the dominant conceptual directions using eigendecomposition. The alignment between natural and synthetic co-occurrence structures is assessed using cosine similarity heatmaps between the top-100 eigenvectors of each co-occurrence matrix. These 100×100 similarity matrices are visualized as square heatmaps where perfect diagonal alignment would indicate identical principal concept axes, while off-diagonal patterns reveal would revolve rotations and mismatches in compositional geometry.

B Custom RA-SAE Training Details

This section summarizes the training details of our custom archetypal SAE (RA-SAE) [56]. The model is open-sourced at a software repository.

Dataset. The auto-encoder is trained on the complete ImageNet-1k training split, (≈ 1.28 M) RGB images. Each image is converted to 261 visual tokens using DINOv2 [135]; tokens are fed to the SAE without class or position embeddings. The total number of training tokens is therefore $50 \times 1.28\text{M} \times 261 \approx 1.67 \times 10^{10}$.

Dictionary. The dictionary has 32,000 concept dimensions. For the sparse activation rule, top- k masking with $k = 5$, is used; activations outside the largest five per input are set to 0. The weights are initialized using Xavier/Glorot. The training is conducted at mixed precision (fp16), with the last ten epochs performed at full precision.

Optimizer and Schedule. The model is trained for 50 epochs using base AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$) optimizer is employed with weight decay set to 10^{-5} . Linear warm-up is applied on the first 5% of steps, followed by cosine decay from $\eta_{\max} = 5 \times 10^{-4}$ to $\eta_{\text{final}} = 10^{-6}$. MSE loss is used alongside an auxiliary term penalizing activations that never enter the top- k set, where $\lambda = 10^{-5}$.

955 C Computational Resources

956 This section summarizes the GPU resources used for training and experiments in support of this
957 paper. In total, we used approximately 202 GPU-hours on NVIDIA H100s and H200s.

958 **RA-SAE.** Trained for approximately 24 GPU-hours on three NVIDIA H100s.

959 **Synthesized Images.** Generating the full $D_{g\theta}$ (see Appendix A) took roughly 5 hours per generator
960 when distributed across four NVIDIA H200 GPUs. With five generators, this totaled approximately
961 100 GPU-hours on a NVIDIA H200.

962 **• Distribution-Level Analysis.** Extracting energy differences at the distribution level took about 3
963 hours per generator on a single NVIDIA H200 GPU (total ~ 15 GPU-hours).

964 **• Datapoint-Level Analysis.** Computing datapoint-level energy differences, ranking concepts per
965 datapoint, and ranking datapoints per concept also took approximately 3 hours per generator on one
966 H200 GPU (total ~ 15 GPU-hours).

967 D Autointerpretability

968 In an effort to automate the interpretation of concepts identified by the RA-SAE, we qualitatively
 969 evaluated the ability of Vision-Language Models (VLMs) to describe them. In particular, we queried
 970 ChatGPT 4o via the OpenAI API using various prompts and formats of an image x'' in which the
 971 given concept was salient, recording the description generated by the VLM. In this section, we
 972 highlight the findings from our exploration. For evaluation purposes, we allowed the VLM to provide
 973 its full reasoning; to make this a scalable solution, a simple modification of the prompt—asking the
 974 VLM to provide only the description—would filter out the reasoning.

975 First, we provided the VLM with two images: a raw version of x'' and x'' with a heatmap superim-
 976 posed to localize the intensity of the concept:

Input



You are given two images: (1) raw image and (2) the same image with a heatmap of areas that are salient for a concept present in the image, as identified by a DINOv2-based SAE. Describe what is this concept.

Output

The concept identified by the DINOv2-based SAE in these images appears to correspond to **"ring"** or **"metal ring"** as the salient object.

Reasoning:

- In the original image, a sleek, brushed-metal ring is centered on a wooden surface.
- In the heatmap overlay, the regions that are most activated (least transparent with bright color blocks) are tightly focused on the ring itself, particularly the outer surface and inner reflections.
- The wooden background is mostly ignored except for minor scattered activations, confirming that the concept is not about the background texture.
- The consistent and strong focus on the metallic, circular object suggests that the model recognizes this as the defining element—likely trained to detect the shape, material, and reflectivity typical of rings.

977 As seen in the example, the VLM did not interpret the heatmap visualization correctly. Instead of
 978 describing the wood texture in the background—the actual salient region—it described the ring in the
 979 foreground, which received no attention from the RA-SAE.

980 To determine whether this was a prompting issue, we tested the same scenario again—this time using
 981 an expanded prompt that explained the heatmap in greater detail:

Input



You are given two images: (1) raw image and (2) the same image with a superimposed heatmap of areas that contain a concept identified by a DINOv2-based SAE. Red areas are the most salient (contain the concept); blue areas are the least salient (do not contain the concept). Describe what is this concept.

Output

The concept identified by the DINOv2-based SAE in these images appears to be **"ring"** or more specifically **"metal band/jewelry ring."**

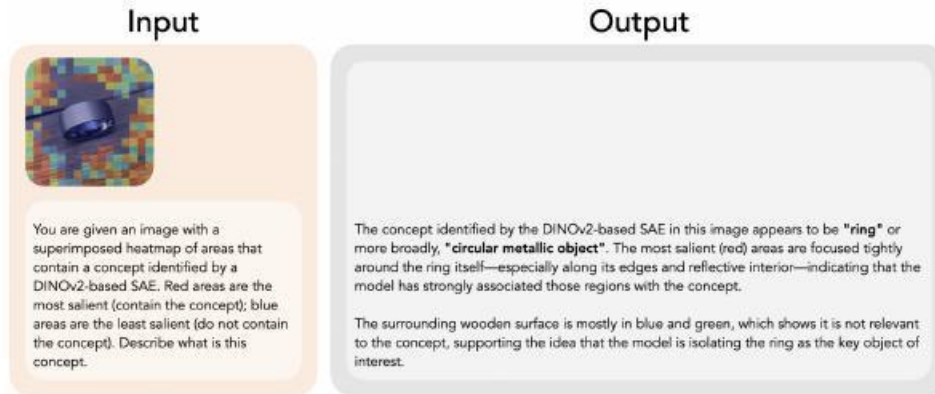
Here's the reasoning:

- In the heatmap image, the red and yellow (most salient) regions align very closely with the circular metal ring at the center of the image.
- The ring is the only distinct object in the photo and is clearly segmented from the background (wooden surface), which is mostly marked in blue and green (least salient).
- The attention does not focus on the texture of the background or lighting but is highly localized to the curved, metallic object.

Thus, the concept is almost certainly the metallic ring—its shape, reflectivity, and material seem to trigger the sparse autoencoder's activation, making it a learned concept in the SAE dictionary.

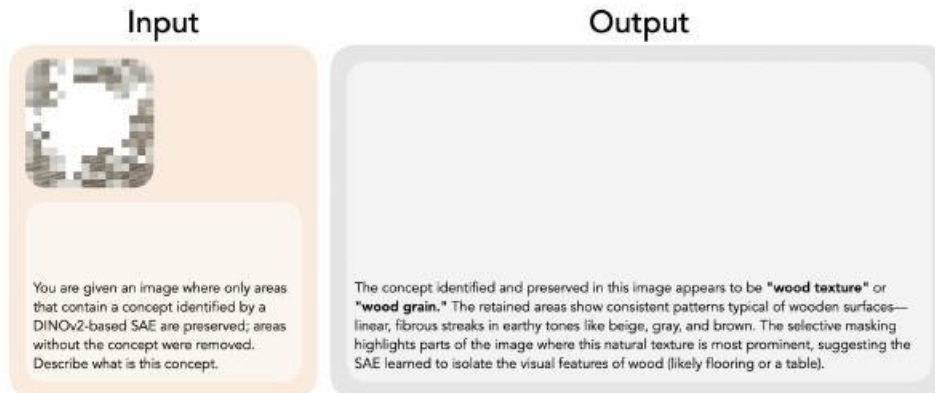
982 This prompt modification did not resolve the issue, as the VLM still identified the ring as the
 983 prominent concept in the image. Several other prompting techniques, including those describing the
 984 heatmap in even greater depth (e.g., by including the code used for its generation), were explored
 985 with similar results.

986 We next tested providing the VLM only with the heatmap-superimposed x'' and a similar prompt
 987 explaining how to interpret the heatmap:



988 Once again, the result failed to identify the wooden background as the prominent concept in the image
 989 and instead focused on the ring. As in the previous case, we experimented with various prompting
 990 strategies using this image format, but the results remained largely unchanged.

991 From these observations, we concluded that the heatmap alone is insufficient to guide the VLM's
 992 attention to the salient areas. The presence of less relevant regions appears to distract the model. To
 993 address this, we replaced the heatmap-superimposed x'' with an alpha-masked version of x'' :



994 As shown in the example, the VLM was now able to correctly identify wood texture as the dominant
 995 concept. We observed consistent success across a range of concepts and images.

996 The formatting of x'' and the prompt shown above yielded the most reliable results in our qualitative
 997 evaluation. However, we note that this evaluation is limited by its qualitative nature (due to the
 998 absence of ground truth annotations) and its focus on a single VLM. We hope future work on the
 999 autointerpretability of SAE concepts can build on and expand this analysis.

1000 E Additional Results: Model-Specific Blindspots



Figure 14: Examples of conceptual blindspots in **Stable Diffusion 1.5**. For each concept, the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots), based on the highest absolute activation, is shown. The spatial heatmap for the concept is superimposed atop the image.

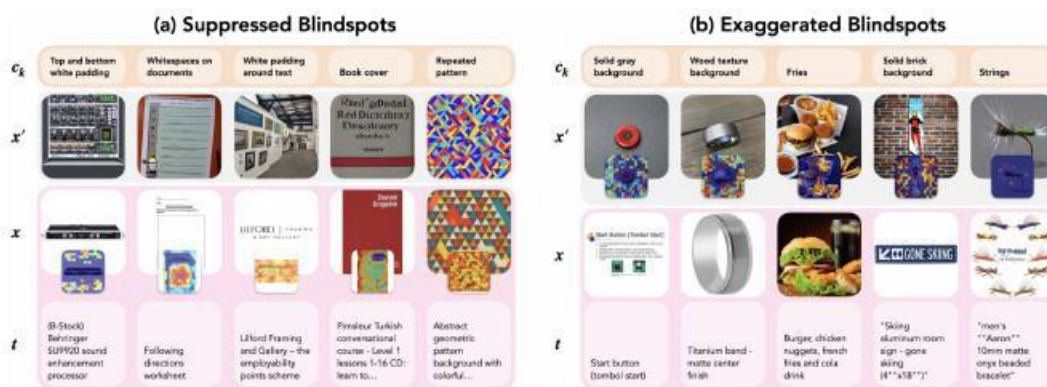


Figure 15: Examples of conceptual blindspots in **Stable Diffusion 2.1**. For each concept, the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots), based on the highest absolute activation, is shown. The spatial heatmap for the concept is superimposed atop the image.

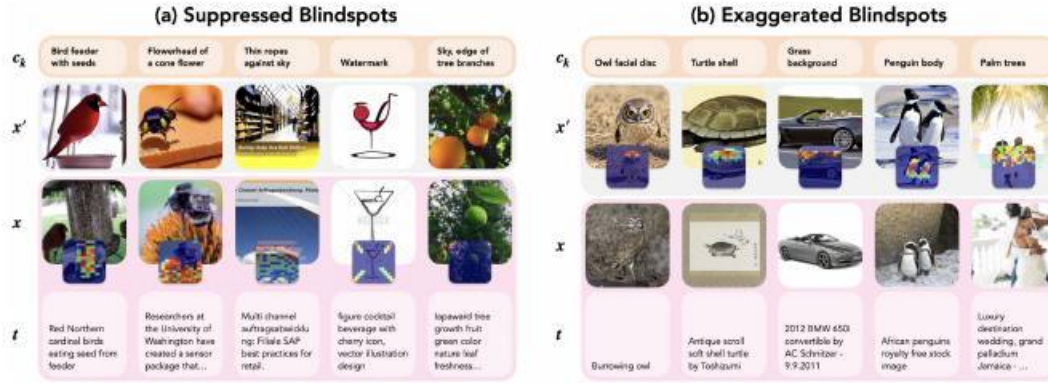


Figure 16: Examples of conceptual blindspots in **Kandinsky**. For each concept, the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots), based on the highest absolute activation, is shown. The spatial heatmap for the concept is superimposed atop the image.



Figure 17: Examples of conceptual blindspots in **PixArt**. For each concept, the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots), based on the highest absolute activation, is shown. The spatial heatmap for the concept is superimposed atop the image.

F Additional Results: Higher-order Blindspots with Compositional Discrepancy

Thus far, our analysis has centered on individual concept activations. Yet visual scenes are rarely composed of isolated concepts; instead, they are structured through rich and structured co-occurrence patterns that encode compositional semantics. We now examine whether generative models capture this higher-order structure by analyzing the co-activation matrix $Z^T Z$, which reflects pairwise correlations in concept usage.

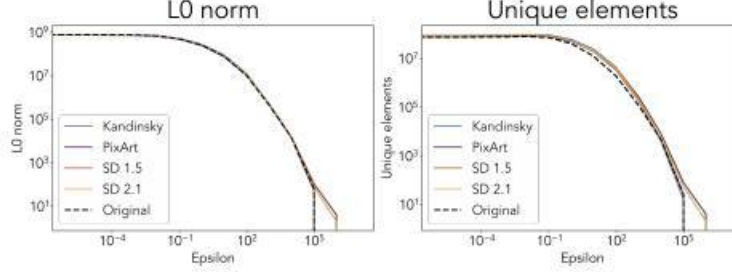


Figure 18: **Sparsity and Structural Divergence.** On the left: L0 norm of the co-occurrence matrix ZZ^T as a function of ϵ (threshold), indicating how many entries remain active in each model. On the right: Number of unique entries in the synthesized distribution relative to the natural distribution. All evaluated models preserve global sparsity structure, but diverge in activation content.

Surprisingly, when assessed at the level of binary structure, diffusion models approximate the global sparsity of the natural co-occurrence matrix with high fidelity. As shown in Fig. 18 (left), the ℓ_0 norm of $Z^T Z$ – thresholded at varying ϵ values – tracks closely between the natural and synthesized distributions across all models. This indicates that the gross connectivity of the conceptual graph, i.e., which concepts tend to co-activate at all, is well preserved. Formally, one can deem $Z^T Z$ as the adjacency matrix of a weighted, undirected graph over concepts, where edge weights reflect co-activation strength across the dataset.

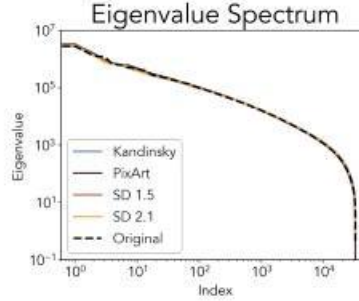


Figure 19: **Spectral Structure of Co-occurrence.** Log-log plot of the eigenvalue spectra from co-occurrence matrices ZZ^T across models and the natural distribution. All evaluated models match the heavy-tailed decay of the natural distribution.

However, as illustrated in Fig. 18 (right), the specific content of these co-activations diverges: a substantial portion of entries in the model-generated $Z^T Z$ are not shared with the natural baseline. This suggests that while the capacity for compositionality is retained, the identity of active pairings may shift, potentially reflecting model specific inductive biases or training artifacts. To probe the internal structure of these co-occurrence patterns, we turn to spectral analysis. Fig. 19 shows the eigenvalue spectra of the co-occurrence matrices for each model and the natural distribution. All spectra exhibit a heavy-tailed decay, consistent with power-law behavior, indicating that generative models preserve the overall rank structure and variance allocation across conceptual dimensions.

Further, we examine the alignment of dominant conceptual directions via cosine similarity heatmaps between the top 100 eigenvectors of the synthesized and natural co-occurrence matrices (Fig. 20). While all models exhibit partial diagonal alignment—implying overlap in principal concept axes—the

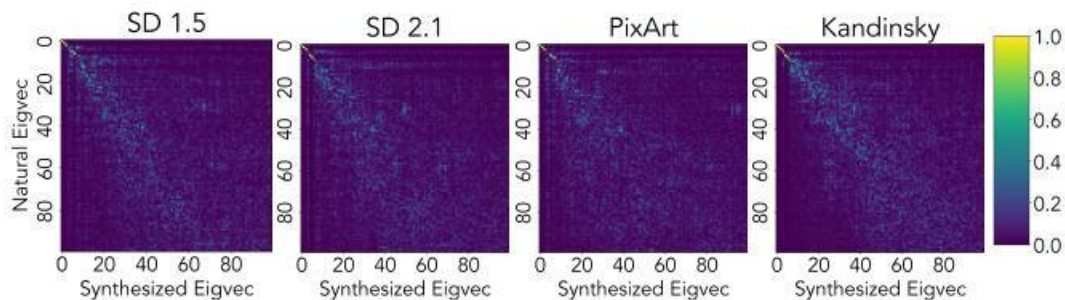


Figure 20: ● **Concept Basis Similarity.** Cosine similarity heatmaps between the top 100 eigenvectors of the natural and synthesized co-occurrence matrices ZZ^T . Diagonal structure shows alignment of dominant conceptual directors, with varying degrees of alignment across the four models.

1027 off-diagonal entries reveal rotations and mismatches in higher modes, reflecting evident deviations in
 1028 compositional geometry.

1029 Together, these findings reveal that diffusion models approximate the global shape of concept co-
 1030 activation surprisingly well, yet deviate in subtle and structured ways when examined through
 1031 the spectral lens. Such higher-order discrepancies may underpin failures in generating coherent,
 1032 multi-object scenes or relational concepts.

1033 G Additional Results: Blindspot Stress Testing

1034 To stress-test the blindspots identified by our method, we gathered a range of prompts describing
 1035 these blindspots and used them to generate many images. We then contrasted the outputs from models
 1036 in which the concept was identified as a blindspot with those in which it was not.

1037 Specifically, ChatGPT-4o was prompted as follows: *I want to generate an image of the following*
 1038 *concept: "<blindspot>". Suggest 50 prompts highlighting this concept to be used as input for a*
 1039 *text-to-image model. Return these as a list of strings in Python.* Five images were generated per
 1040 prompt and analyzed using our custom RA-SAE model (see Appendix B), which ranked them by
 1041 the intensity with which the desired concept appeared. All images were then manually reviewed to
 1042 determine whether the blindspot was successfully depicted.

1043 As seen in the following examples of suppressed concepts, while some aspects of the target concept
 1044 occasionally appeared (e.g., a holder or string for the bird feeder blindspot and a round hole for
 1045 the glossy DVD disc blindspot), the models generally failed to generate the full concept. This
 1046 aligns with our method’s assessment and supports the validity of the stress test.

1047 G.1 Bird Feeder Blindspot in Kandinsky



Figure 21: Examples of images generated with various prompts involving the bird feeder concept as a part of the stress testing. In Kandinsky, our method identified this concept as a suppressed conceptual blindspot, which matches the observed behavior: the model is unable to generate a corresponding image. By contrast, SD 1.5, in which this concept was not identified as a blindspot, is able to generate this concept.



Figure 22: Examples of images generated with various prompts involving the glossy DVD disc concept as a part of the stress testing. In SD 1.5, our method identified this concept as a suppressed conceptual blindspot, which matches the observed behavior: the model is unable to generate a corresponding image. By contrast, Kandinsky, in which this concept was not identified as a blindspot, is able to generate this concept.

1049 **H Additional Examples: Synthesized Images**



Figure 23: Additional image-caption pair examples from LAION-5B with matching images generated with the same prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

1050 I Concentration bounds for δ

1051 In our experiments, we estimate $\delta(k)$ using $n = 10,000$ paired samples for each concept. While this
 1052 budget is modest, it raises the natural question of whether it suffices to obtain reliable estimates. To
 1053 address this, we derive a concentration bound on the empirical estimator $\hat{\delta}_n(k)$ using McDiarmid’s
 1054 inequality [142]. The resulting bound is tight and demonstrates that even with relatively few samples,
 1055 we can obtain fast and accurate estimates of concept bias.

1056 **Theorem 4** (Concentration of $\hat{\delta}_n(k)$). *We assume that the concept score $\xi_k(\mathbf{x})$ takes values in $[a, b]$
 1057 almost surely for all images $\mathbf{x}$ drawn from either $\mathcal{D}_X$ or $\mathcal{D}'_X$. Let n paired samples $(\mathbf{x}_i, \mathbf{x}'_i)_{i=1}^n$ be
 1058 drawn independently with $\mathbf{x}_i \sim \mathcal{D}_X$ and $\mathbf{x}'_i \sim \mathcal{D}'_X$, and define the empirical estimator*

$$\hat{\delta}_n(k) := \sigma \left(\frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}_i) - \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}'_i) \right).$$

1059 Let $M := b - a$ and $L := M/4$. Then for every $\varepsilon > 0$, the deviation satisfies

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right).$$

1060 *Proof.* The function $x \mapsto \sigma(x)$ is $1/4$ -Lipschitz, since $|\sigma'(x)| \leq 1/4$ for all x .

1061 Viewing $\hat{\delta}_n(k)$ as a function of the $2n$ independent variables $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n)$, changing a
 1062 single argument alters the inner difference of means by at most M/n , and the outer sigmoid scales this
 1063 by at most $1/4$. Hence, the bounded difference constant for each coordinate is $(M/n)(1/4) = L/n$.

1064 By McDiarmid’s inequality [142],

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{j=1}^{2n} (L/n)^2} \right) = 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right),$$

1065 which proves the claim. □

1066 Practically, most concept scores $\delta(k)$ are sparse, with the majority concentrated near zero and only
 1067 a few reaching values up to 10. The concentration bound shows that even for the largest observed
 1068 biases, a sample size of $n = 10,000$ yields estimates of $\hat{\delta}_n(k)$ that deviate from the true value by
 1069 no more than a small ε with high probability. This justifies our sampling strategy and confirms that
 1070 accurate bias measurements are attainable with limited data.

J Monotonicity and Calibration-Free Interpretation of δ

Our goal when analyzing blind spots is to rank concepts by the severity of their generative bias. In practice, we use the score $\delta(k)$ for this purpose. However, one may wonder whether such a score introduces distortions relative to more direct quantities such as the energy gap or the odds ratio. The following result establishes that $\delta(k)$ is a strictly increasing reparameterization of both, and therefore inherits their ordering. This guarantees that no calibration is needed when using $\delta(k)$ to rank concepts.

Theorem 5 (Monotonicity and Calibration of $\delta_{g_\theta \leftrightarrow G}$). *For every concept index k define the energy gap*

$$\Delta_k = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}'_{\mathcal{X}}}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{\mathcal{X}}}[\xi_k(\mathbf{x})],$$

the associated odds ratio $\rho_k = \exp(\Delta_k)$, and the energy-difference score

$$\delta(k) = \frac{1}{1 + \exp(-\Delta_k)} = \frac{\rho_k}{1 + \rho_k}.$$

Then $\delta(k)$ is a strictly increasing bijection of both Δ_k and ρ_k , so ranking concepts by any one of $\delta(k)$, Δ_k , or ρ_k produces exactly the same ordering.

Proof. The logistic sigmoid satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x)) > 0, \forall x \in \mathbb{R}$; hence σ and therefore $\delta(k) = \sigma(\Delta_k)$ grow strictly with Δ_k . Because the exponential map is also strictly increasing and bijective $\mathbb{R} \rightarrow (0, \infty)$, setting $\rho_k = \exp(\Delta_k)$ preserves order and gives $\Delta_k = \log \rho_k$. Substituting this identity into σ yields $\delta(k) = \sigma(\log \rho_k) = \rho_k / (1 + \rho_k)$, which is the composition of two strictly increasing bijections and is therefore itself strictly increasing and bijective in ρ_k . Since strict monotonic functions never reverse inequalities, the three quantities share the same total order over concepts.

□

Thus, ranking concepts by $\delta(\cdot)$ is strictly equivalent to ranking them by energy gap or by conceptual generation odds ρ_k . No calibration is necessary, and all three quantities preserve the same total ordering over concepts.

1094 K Stability of FID Under SAE Embeddings

1095 In this section we establish a quantitative relationship between the Fréchet Inception Distance
1096 (FID) computed in the original activation space (of dimension d) and the FID after applying a
1097 (potentially overcomplete) SAE dictionary $D \in \mathbb{R}^{k \times d}$ with $k \gg d$. Throughout we assume that D
1098 has orthonormal columns but is not necessarily square, i.e.

$$D^\top D = I_d, \quad \text{while} \quad DD^\top \neq I_k.$$

1099 We start by recalling a simple fact: if D is not overcomplete, orthogonal and $k = d$, then we have an
1100 isometry between A and Z , implying that the FID is perfectly preserved. However, this case is not
1101 realistic, we will then turn to the overcomplete case, and show we can bound FID by the extreme
1102 singular value of D . We will work with the Wasserstein-2 metric $\mathcal{W}_2$, noting that FID is just $\mathcal{W}_2^2$
1103 specialised to Gaussians.

1104 For a probability measure μ on $\mathbb{R}^d$ we write $D_\# \mu$ for its push-forward under D , i.e. $D_\# \mu(z) =$
1105 $\mu(D^{-1}z)$. Denote by $\sigma_{\min}$ and $\sigma_{\max}$ the minimal and maximal singular values of D , equivalently
1106 the square-roots of the extremal eigenvalues of $DD^\top$:

$$\sigma_{\min}^2 I_k \preceq DD^\top \preceq \sigma_{\max}^2 I_k.$$

1107 Empirically one usually finds $\sigma_{\min}, \sigma_{\max} \approx 1$, but the proof does not rely on that. We will start by a
1108 simple lemma in the case where D is not overcomplete.

1109 **Lemma 1** (Isometry under exact orthogonality). *Suppose $k = d$ and $D^\top D = DD^\top = I_d$. Then*
1110 *D is an isometry: $\|Dv\|_2 = \|v\|_2$ for all $v \in \mathbb{R}^d$. Consequently, for any probability measures μ, ν*
1111 *on $\mathbb{R}^d$ with finite second moment,*

$$\mathcal{W}_2(D_\# \mu, D_\# \nu) = \mathcal{W}_2(\mu, \nu).$$

1112 *Proof.* Orthogonality of D implies preservation of the Euclidean norm, and push-forward commutes
1113 with the map inside the $\mathcal{W}_2$ infimum; the integrand is unchanged, so the infimum value is identical.
1114 □

1115 This case, however, is quite unrealistic as SAE usually rely on the overcompleteness to extract
1116 meaningful and interpretable concepts. In the overcomplete case, D is no longer orthonormal, but
1117 we can still have column-orthonormal dictionary. We will use that to show that we can bound using
1118 the extremal singular value of $D^\top D$.

1119 **Theorem 6** (FID under column orthogonal embeddings). *Let $D \in \mathbb{R}^{k \times d}$ satisfy $D^\top D = I_d$ and*
1120 *denote by $0 < \sigma_{\min} \leq \sigma_{\max}$ the extreme singular values of $DD^\top$. Given two data matrices*
1121 *$A, A' \in \mathbb{R}^{n \times d}$ (rows are sample vectors), set $Z = AD^\top \in \mathbb{R}^{n \times k}$ and $Z' = A'D^\top \in \mathbb{R}^{n \times k}$. Then*

$$\sigma_{\min}^2 \text{FID}(A, A') \leq \text{FID}(Z, Z') \leq \sigma_{\max}^2 \text{FID}(A, A').$$

1122 *Proof.* Write μ for the empirical measure of A and ν for that of A' , i.e.

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{A_{i,:}}, \quad \nu = \frac{1}{n} \sum_{i=1}^n \delta_{A'_{i,:}}.$$

1123 For any coupling $\pi \in \Pi(\mu, \nu)$ (i.e. a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ, ν) we have,
1124 by the extremal singular value bound,

$$\sigma_{\min}^2 \|x - y\|_2^2 \leq \|D(x - y)\|_2^2 \leq \sigma_{\max}^2 \|x - y\|_2^2, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d.$$

1125 Integrating with respect to an arbitrary coupling $\pi \in \Pi(\mu, \nu)$ yields

$$\sigma_{\min}^2 \int \|x - y\|_2^2 d\pi \leq \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \int \|x - y\|_2^2 d\pi.$$

1126 The middle integral is exactly the transport cost of the pushed-forward coupling $(D \times D)_\# \pi$ between
1127 $\mu_D := D_\# \mu$ and $\nu_D := D_\# \nu$. Because the inequalities hold for every π , they hold in particular

1128 for the optimal couplings attaining $\mathcal{W}_2(\boldsymbol{\mu}, \boldsymbol{\nu})$ and $\mathcal{W}_2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D)$, though these two optima need not
 1129 coincide. Taking the infimum over $\boldsymbol{\pi}$ term-wise makes this explicit:

$$\sigma_{\min}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi} \leq \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|D(\mathbf{x} - \mathbf{y})\|_2^2 d\boldsymbol{\pi} \leq \sigma_{\max}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi}.$$

1130 Hence

$$\sigma_{\min}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}) \leq \mathcal{W}_2^2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D) \leq \sigma_{\max}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

1131 Recognising $\text{FID}(\cdot, \cdot) = \mathcal{W}_2^2(\cdot, \cdot)$ for the Gaussian surrogate and plugging in $(\mathbf{A}, \mathbf{A}')$ (resp. $(\mathbf{Z}, \mathbf{Z}')$)
 1132 finishes the proof. $\square$

1133 Essentially, theorem 6 tells us that applying a column-orthogonal overcomplete SAE dictionary
 1134 cannot distort Fréchet Inception Distance by more than the square of its extremal singular values.
 1135 When $\mathbf{D}$ is nearly orthogonal – empirically we usually found that $\sigma_{\min}, \sigma_{\max} \approx 1$ – the result implies
 1136 that FID measured in the SAE feature space is essentially close to the canonical FID. $\square$

1137